

Design Specification: PARD Programming Language

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1 Introduction & Philosophy

PARD is a programming language designed around game design and development. It leverages FORTRAN's fast math and C interoperability. The name PARD is an abbreviation of Punching Card, which is an homage to the early days of FORTRAN.

2 Lexical Grammar

2.1 Variables and their types

Variables, just like most other programming languages, are a significant part of PARD. There's 4 basic variable types:

- **int**: A standard integer type for counts, loop counters, and IDs.
- **float**: A floating-point number for positions, physics calculations, and deltatime.
- **bool**: A boolean value (**true** or **false**) for state flags.
- **string[N]**: A fixed-length, ultra-fast character array of size N .

As you can see, strings are not dynamically sized - this is not only for easy operation in fortran, but it also introduces a good performance boost.

2.2 Variable definition

You can easily define a variable in the following way:

```
int myInt = 10
float myFloat = 10.5
bool myBool = true
string[6] myString = 'Hello!'
```

3 Compiler Architecture